

FR-HHT Locking & Chirp — Interim Update

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Executive summary

HHT IMF2 locking: baseline axis $\ln(1+z)$ ($\alpha=1.0$) outperforms FR-style axis ($\alpha=0.5$) on $|r|$; p-values tied at 0.00498 for the tested pairs.

FR SN+BAO (spline Λ): allowing $c(z)$ in BAO ('c_of_z') gives a modest χ^2 improvement vs constant-c in BAO ('c0').

Piecewise- Λ 'discrete events': $\Delta z \rightarrow 1e-4$ reproduces the continuous result within rounding; coarse $\Delta z=0.015$ is slightly worse (as expected).

OU $\Lambda(z)$ sensitivity: results stable from $\text{amp}=1e-120 \dots 1e-70$; degrade clearly by $\text{amp} \approx 1e-2$. Corr-length sweep (0.05–0.60) showed no material change.

PBH $\Lambda \leftrightarrow A_{\text{acc}}$ coupling $\gamma \in \{0, 0.02, 0.05, 0.10\}$: no change in best-fit or χ^2 with SN-only — suggests the tested γ range is too small or SN-only lacks constraining power.

Soft-link (energy–luminosity) with flat FR and $\log_{10} p_z$ scalings: initial $\kappa = \{-1.0, -0.75\}$, $\sigma=0.10$ runs converge to identical solutions; likely penalty not binding at this σ .

HHT IMF2 locking ($\alpha=0.5$ vs $\alpha=1.0$)

Setup: ceemdan, IMF=2, packs=bao_hht_2bin_z0p7_longbest, seeds={3,7,11}, ns=200.

Summary table: mean $|r| \rightarrow \alpha=0.5$: 0.8790809752, $\alpha=1.0$: 0.9093863675; median p both 0.0049751244; win rate ($\alpha=0.5$) = 0/3.

FR (spline Λ) with SN+BAO — BAO c-mode comparison

c0: $\chi^2=2017.2954$, $\chi^2/\text{dof}=1.18247$, AIC=2031.2954, BIC=2069.4174, $H_0=59.576$, $\Omega_m=0.3435$, $\Omega_k=0.00779$, $w_{\text{eff}} \approx -0.912$.

c_of_z: $\chi^2=2014.5488$, $\chi^2/\text{dof}=1.18086$, AIC=2028.5488, BIC=2066.6709, $H_0=60.579$, $\Omega_m=0.1333$, $\Omega_k=0.25117$, $w_{\text{eff}} \approx -1.080$.

'c_of_z' gains ~ 2.75 in χ^2 vs 'c0' with these settings; parameters shift toward higher $|\Omega_k|$ and more negative w_{eff} .

Piecewise- Λ : continuous vs discrete-events

continuous: $\chi^2=1975.2424$, AIC=1989.2424 (baseline)

discrete $\Delta z=0.015$: $\chi^2=1980.4284$, AIC=1994.4284 (coarser events worse)

discrete $\Delta z=1e-4$: $\chi^2=1975.2100$, AIC=1989.2100 (matches continuous within rounding)

OU $\Lambda(z)$ sensitivity

Correlation-length sweep (amp=1e-120): $\text{corr}_z \in \{0.05, 0.15, 0.30, 0.60\} \rightarrow$ identical best-fit & $\chi^2=1974.8730$ (SN-only).

Amplitude sweep at fixed $\text{corr}_z=0.30$ and fixed parameters ('sampler=none'): 1e-120...1e-70 unchanged; 1e-2 degrades ($\chi^2 \approx 2089.88$).

Probe MCMC at amp=1e-2 (short chain): $\chi^2 \approx 2020.38$.

PBH $\Lambda \leftrightarrow A_{\text{acc}}$ coupling (γ) — SN-only

$\gamma \in \{0.00, 0.02, 0.05, 0.10\} \rightarrow \chi^2=1974.5117$ for all; no effect across this range for SN-only.

Action: include BAO/Planck and/or widen γ grid.

Soft-link experiments (flat FR, log1pz scalings)

$\kappa=-1.0, \sigma=0.10 \rightarrow \chi^2=2005.9505, \text{AIC}=2017.9505$;

$\kappa=-0.75, \sigma=0.10 \rightarrow \chi^2=2005.9505, \text{AIC}=2017.9505$;

Identical outcomes suggest the soft constraint isn't binding at $\sigma=0.10$. Action: repeat for $\sigma \in \{0.05, 0.02, 0.01\}$.

Data & environment notes

Planck prior: 'planck_distance_prior.json' missing in /Users/boyde/Downloads/ — add file to enable --planck and --planck-c-mode tests.

emcee: use nwalkers ≥ 14 for 7 parameters; corner is optional.

HHT α -compare DOCX needed python-docx (installed).

Recommended next steps

HHT: $\alpha \in \{0.6 \dots 1.4\}$ (step 0.1) for IMF2/IMF3; keep surrogate trials; report $|r|$ and p-value distributions per pad/method.

BAO/Planck: once Planck JSON present, compare c_0 vs c_{of_z} under Planck distance priors; inspect Ω_k and w_{eff} .

Discrete events: quantify convergence vs Δz ladder (1e-3, 5e-4, 1e-4) with AIC/BIC.

OU $\Lambda(z)$: place informative priors on corr_z and amplitude; test rejection thresholds for large amplitudes ($\geq 1e-4$).

PBH coupling: extend γ grid and include BAO/Planck; track posteriors of A_{acc} , n_{acc} , and γ .

Soft-link: sweep $\kappa \in [-1.5, +1.5]$ with $\sigma \in \{0.05, 0.02, 0.01\}$; ensure $n_{\text{steps}} \geq 2500$.

Repro command snippets

```
# FR + SN+BAO (spline  $\Lambda$ ):  $c_{\text{of\_z}}$ 
%run '/Users/boyde/.spyder-py3/PLamb_Test_10V6c_plus.py' \
--model FR --data '/Users/boyde/Downloads/Pantheon+SH0ES.dat' \
--cov '/Users/boyde/Downloads/Pantheon+SH0ES_STAT+SYS.cov' \
--bao '/Users/boyde/Downloads/bao_long.csv' --rd 147.09 \
--sampler emcee --nwalkers 24 --nsteps 800 --nburn 200 --nint 220 --seed 73 \
--fluct-Lambda --Lambda-model spline --Lambda-amp 1.0e-120 --Lambda-knots 7 \
--bao-c-mode  $c_{\text{of\_z}}$  --out 'plamb_runs/SNBAO_baocmode_ $c_{\text{of\_z}}$ '

# Piecewise- $\Lambda$ :  $\Delta z \rightarrow 1e-4$  matches continuous
%run /Users/boyde/.spyder-py3/PLamb_Test_10V6c_plus.py \
--model FR --data /Users/boyde/Downloads/Pantheon+SH0ES.dat \
--cov /Users/boyde/Downloads/Pantheon+SH0ES_STAT+SYS.cov \
--sampler none --nint 400 \
--H0 82.5742 --Om 0.68207 --Ol 0.96443 --A_acc -0.25624 --n_acc -5.84476 \
--gamma_c -0.02980 --epsilon_M 0.38324 \
--fluct-Lambda --Lambda-model piecewise --Lambda-amp 1e-120 --Lambda-knots 8 \
--discrete-events --event-size 0.0001 \
--out 'plamb_runs/SN_piecewise_disc_at_cont_DZ1e-4'
```